Smart IT Support Assistant

Jordan University of Science and Technology (JUST)

*An LLM-Powered IT Support Assistant for JUST Students

Phase 1: Problem and Data

1. Team and Roles:

Team Members	ID	Role
Rand Al-Darabseh	174269	Product Lead
Batool Abualhayjaa	168361	Data Lead
Hiba Al-Naimat	170251	Evaluation Lead & Engineering Lead

2. User and Problem:

Target Users:

The target users of the Smart IT Support Assistant are students at Jordan University of Science and Technology (JUST) who experience technical difficulties with university services and personal devices.

Problem:

Students often face technical issues related to Wi-Fi, university platforms, and accessing IT services. Finding accurate solutions can be time-consuming, as students rely on online searches, AI tools, classmates, or university IT support.

Current Solution:

Currently, students try to solve technical problems themselves, search online, ask classmates, or contact university IT support. However, these methods may not always provide quick or reliable solutions.

Evidence of the Problem:

A survey was conducted with 41 respondents, including 39 JUST students. The results showed that 58.5% experienced Wi-Fi or internet issues, 51.2% faced e-learning platform problems, and 70.7% reported that technical issues affected their studies or assignments. Additionally, 82.9% expressed willingness to use an AI-powered IT support assistant. These findings highlight the need for an accessible and reliable IT support solution.

C:\Users\heban\Downloads\just survey result .pdf

3. What the System Will Do: The Smart IT Support Assistant will help students at Jordan University of Science and Technology (JUST) solve common technical problems related to university IT services and personal devices. The system will use Large Language Model (LLM) and Retrieval-Augmented Generation (RAG) technologies to provide accurate, clear, and reliable answers. When a student submits a question in natural language, the system will retrieve relevant information from official university IT documents, guidelines, and trusted technical documentation. Based on the retrieved If the required information is not available in the retrieved documents, the assistant will inform the student that it cannot provide a verified answer rather than generating unsupported information.

The following examples illustrate possible interactions between the user and the system:

User Input	System Answer
How can I connect to the university Wi-Fi?	The assistant provides the official Wi-Fi connection instructions and explains the required steps.
What should I do if I cannot access JUST Learn?	The assistant retrieves the relevant troubleshooting instructions and provides steps to resolve the login issue, along with the official source.

4. Data: The project will use official IT-related documents and resources published by Jordan University of Science and Technology (JUST), including student IT support instructions, user guides, troubleshooting procedures, and information about university IT services such as student accounts, university email, Wi-Fi connectivity, and the e-learning platform.

The main data sources will be the official JUST Information Technology and Communications Center (ITCC), JUSTLearn – Center for E-Learning and Open Educational Resources, and the JUST Student Services Portal.

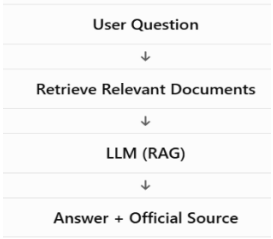
The team will collect and review relevant documents from these sources to ensure the information is accurate, reliable, and suitable for answering common student IT support questions.

Data Sources:

1. JUST IT & Communications Center (ITCC)

2. JUSTLearn – Student Support

5. Planned Architecture: The system will use RAG to retrieve relevant information from trusted IT support documents. An LLM will then generate clear troubleshooting instructions with source references.



7. How We Will Measure Success:

Success will be measured by answer accuracy, source reliability, and hallucination prevention.

8. Risks:

Risk	Mitigation
Missing documents	Find alternative sources.
Incorrect answers Limited time	Provide citations. Prioritize essential services.